

The Asura Parallel Empire Codex v1.0

Sovereign Founding Blueprint for the Post-Indra Age (2026–2031)

“The Asura does not storm the gates of Amaravati. The Asura builds a civilization so complete that Amaravati becomes optional.”

“ $N = \sum_i d_i \cdot 3^i$ where $d_i \in \{-1, 0, 1\}$. Every integer has a unique representation. The sign is the leading digit. There is no sign bit. There is no complement. There is no overhead. This is how sovereign systems are built — by eliminating the need for external correction at the foundational layer.”

Document Control

Field	Value
Version	1.0

Classification	Founding Document — Eyes of the Builder Only
Time Horizon	2026–2031 (5-year primary arc)
Activation Threshold	1 human, 72 hours, \$0 capital required for Phase 0
Mathematical Foundation	Balanced Ternary Arithmetic, Qutrit Quantum Architecture, Eisenstein Integer Geometry
Strategic Precedents	Bitcoin (2009), Linux (1991), Sangha (~500 BCE), Ethereum (2015), Signal (2014)
Last Updated	February 2026

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Part 0: Preamble — The Churning

The Mythic Frame

In the Samudra Manthan, Devas and Asuras both grasped the serpent Vasuki and churned the Ocean of Milk. The poison Halahala emerged first. Then the nectar Amrita. The strategic lesson is exact: the churning requires both polarities, the poison precedes the nectar, and sovereignty belongs to whoever can metabolize the poison and keep churning.

The current technological order — dominated by vertically integrated empires (xAI/Tesla/SpaceX/Starlink/X/Neuralink and their equivalents) — is Indra's Amaravati: impressive, real, consolidated. It is not evil. It is simply *centralized*. It represents one pole of the churn.

This Codex does not prescribe war with Indra. It prescribes the construction of the other pole: a sovereign, compounding, parallel civilization layer that is independent across six critical dimensions — AI, Energy, Truth/Verification, Economy, Governance, and Narrative. The strategy is asymmetric by design. Where Indra builds vertically integrated empires requiring billions in capital, the Asura builds horizontally composable protocols requiring mathematical elegance and network effects.

The Core Thesis

The highest-leverage asymmetric move available in 2026 is to build a sovereign technology stack rooted in balanced ternary mathematics — not because ternary is marginally more efficient than binary, but because it represents a genuinely different computational substrate that unlocks architectural choices unavailable to the incumbent order.

This is not metaphor. It is provable mathematics and engineering reality, as documented in the foundational research that accompanies this Codex.

Why This Artifact Exists

A single founder (or founding team of 2–5) cannot build all six pillars simultaneously. The flywheel must start somewhere. This Codex identifies the single highest-leverage entry point — a ternary-native AI inference engine — and shows how it cascades into the remaining five pillars through compounding network effects, economic gravitational pull, and narrative inevitability.

Part I: The Mathematical Foundation

Why Ternary Is the Substrate of Sovereignty

The mathematical case for balanced ternary is not speculative. It rests on proven theorems, exact identities, and verified engineering results spanning seven decades. What follows is the rigorous

foundation upon which every strategic layer of this Codex is built.

1.1 Radix Economy: Base 3 Is Provably Optimal Among Integers

The cost of representing a number N in base b is approximately:

$$E(b, N) \approx b \cdot \ln(N) / \ln(b)$$

Minimizing $b/\ln(b)$ yields $b = e \approx 2.718$ (Steiner’s problem, 1850). Evaluating at integer bases:

Base	$b/\ln(b)$	Deviation from Optimal
e (theoretical)	2.718	0%
3	2.731	0.46%
2	2.885	6.15%
4	2.885	6.15%

Base 3 is within 0.46% of the theoretical optimum. Base 2 is 6.15% worse. This is not a marginal difference — it compounds across every register, every memory cell, every wire in every chip. The original engineering analysis in *High-Speed Computing Devices* (Engineering Research Associates, 1950) computed that radix-3 registers covering numbers to 10^6 required 38.24 triodes versus binary’s 39.20 — and that was before the discovery of balanced ternary’s additional algebraic advantages.

Strategic implication: Every binary system pays a permanent 6.15% radix economy tax. At civilizational scale, across trillions of computations per second, this tax is enormous. Building

natively in ternary eliminates it.

1.2 Balanced Ternary: The Algebraic Elegance

A number N in balanced ternary is expressed as:

$$N = \sum_i d_i \cdot 3^i \text{ where } d_i \in \{-1, 0, 1\}$$

Every integer has a unique canonical representation. The properties that matter for system-building:

Negation is free. Swap every $-1 \leftrightarrow 1$, leave 0s unchanged. No sign bit. No two's complement. No overflow edge cases. This alone eliminates an entire class of hardware complexity.

Multiplication is carry-free. The single-trit multiplication table: multiplying by -1 is negation, by 0 is zero, by 1 is identity. Zero carries in all nine entries.

Addition has minimal carries. Only 2 of 9 entries in the addition table produce carries ($1+1$ and $(-1)+(-1)$). Compare to binary where 1 of 4 entries carries — same ratio, but balanced ternary's carries are symmetric and self-correcting.

Truncation equals rounding. Because the maximum fractional value in balanced ternary is $\sum_{k=1}^{\infty} 1 \cdot 3^{-k} = 1/2$, the fractional part always lies in $[-1/2, 1/2]$. Truncation automatically selects the nearest integer. Donald Knuth identified this as one of balanced ternary's most elegant properties.

Strategic implication: These aren't theoretical curiosities. They translate directly to simpler hardware, fewer error modes, and faster arithmetic. Microsoft Research's BitNet b1.58 has already demonstrated that ternary weights $\{-1, 0, +1\}$ — encoding $\log_2(3) \approx 1.58$ bits per weight — can match full-precision model accuracy while cutting energy use up to 3×.

1.3 The Fractional Part Engine and Centering Shift

The conversion from standard base-3 digits $\{0, 1, 2\}$ to balanced ternary digits $\{-1, 0, 1\}$ operates through the symmetric residue system mod 3: $\text{sym_mod}(n, 3) = n - 3 \cdot \text{round}(n/3)$. The fractional part function $\{x\} = x - \lfloor x \rfloor$ drives digit extraction in any base. For base 3, the k -th digit is $d_k = \lfloor 3 \cdot \{3^{k-1} \cdot x\} \rfloor$, and the tripling map $T(x) = 3x \bmod 1$ shifts the expansion one position left.

The centering shift $\{x\} - 1/3$ maps the three standard ternary digit intervals onto balanced ternary:

Standard interval	Shift result	Balanced digit
$[0, 1/3)$	$[-1/3, 0)$	-1
$[1/3, 2/3)$	$[0, 1/3)$	0
$[2/3, 1)$	$[1/3, 2/3)$	+1

The total cumulative bias is $\sum_{k=1}^{\infty} 3^{-k} = 1/2$, and $1/3$ is the per-digit contribution. This centering operation is exact, algebraically clean, and directly implementable in hardware.

Hermite's Identity (specialized to $n = 3$):

$$\{x\} + \{x + 1/3\} + \{x + 2/3\} = \{3x\} + 1$$

This identity encodes the ternary structure of the real line: the sum of fractional parts at three equally-spaced shifts equals the fractional part of the tripled argument plus 1. It is periodic with period $1/3$ and directly governs balanced ternary digit extraction.

1.4 The π Obstruction and the π^2 Bridge

The relationship between π and base-3 contains a proven impossibility and a remarkable exception.

The impossibility: Borwein, Galway, and Borwein (2004) proved that no degree-1 BBP-type formula for π exists except in bases that are powers of 2. In base 3, π is always entangled with other constants (arctangent and logarithm terms) that cannot be isolated. This is not a failure of effort — it reflects the special role of powers of 2 in polylogarithm identities at roots of unity.

The exception: π^2 *does* have a base-3 BBP formula (Formula 76 in Bailey's Compendium, updated April 2023). The squaring operation "untangles" π from the obstructions that prevent a base-3 formula for π itself. Under Bailey's "Hypothesis A," any constant with a BBP formula is either rational or normal to the corresponding base — meaning π^2 may be provably normal in base 3, even though π 's normality in any base remains open.

Strategic implication: This mathematical structure suggests that ternary arithmetic has a privileged but non-obvious relationship to fundamental constants. The system we build doesn't need π directly — it needs the algebraic structures that π inhabits, and those structures are more accessible in base 3 through π^2 .

1.5 The Eisenstein Bridge: Where Ternary Meets Deep Number Theory

The Eisenstein integers $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3}$, form a hexagonal lattice in the complex plane. This is where balanced ternary arithmetic and higher geometry genuinely intersect:

The prime 3 ramifies in $\mathbb{Z}[\omega]$ as $3 = -\omega^2(1 - \omega)^2$, making the Eisenstein prime $\lambda = 1 - \omega$ the fundamental building block. The norm $N(a + b\omega) = a^2 - ab + b^2$ is a quadratic form intimately connected to hexagonal lattice geometry.

Euler’s proof of Fermat’s Last Theorem for $n = 3$ depends on the factorization $x^3 + y^3 = (x + y)(x + \omega y)(x + \omega^2 y)$ in $\mathbb{Z}[\omega]$ — ternary structure is literally embedded in one of the deepest results in number theory.

For post-quantum cryptography: The Shortest Vector Problem (SVP) and Learning with Errors (LWE) — foundations of NIST’s standardized post-quantum algorithms Kyber and Dilithium (2024) — depend on lattice geometry. The balanced ternary lattice $\{-1, 0, 1\}^n$ is a natural search space for short vectors. Building cryptographic primitives natively in ternary isn’t aesthetic preference — it’s structural alignment with the mathematics of post-quantum security.

1.6 Qutrits: The Convergence Point

The most promising site for unification of ternary arithmetic, transcendental constants, and quantum computation is the qutrit — a quantum system with three basis states $|0\rangle, |1\rangle, |2\rangle$.

A qutrit lives in $\mathbb{C}^3$ with general state $\alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle$ where $|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1$. Single-qutrit gates are 3×3 unitary matrices from $U(3)$, parameterized by 8 generators — the Gell-Mann matrices of $su(3)$, the same Lie algebra governing the strong nuclear force in particle physics.

The qutrit quantum Fourier transform uses phases $e^{2\pi i/3^n}$, embedding π directly into base-3 arithmetic. This is where the π obstruction becomes an asset: the phases that classical base-3 cannot cleanly extract become native operations in quantum base-3.

Verified experimental results:

Result	Source	Significance
Superconducting qutrit entangling gates	Goss et al., <i>Nature Communications</i> , 2022	Hardware viability proven

First qutrit error mitigation	<i>npj Quantum Information</i> , 2024	Error correction pathway exists
Logarithmic-depth CCNOT decomposition	Gokhale et al., ISCA 2019	Up to 70× gate count improvement over qubits
Counterdiabatic qutrit optimization	<i>Nature</i> , 2025	Up to 90× solution quality improvement over qubits

Strategic implication: Qutrit quantum computing is not speculative — it is being built now, with published results showing order-of-magnitude advantages. A ternary-native classical stack is the natural bridge to qutrit quantum hardware. Building it now creates the software ecosystem that qutrit hardware will need.

1.7 The Engineering Reality: 2025–2026 State of the Art

The hardware is catching up to the mathematics:

Development	Date	Entity	Detail
First energy-efficient ternary MOS (8-inch wafer)	2019	Samsung/UNIST	Published in <i>Nature Electronics</i>
Ternary logic gate patent CN119652311A	March 2025	Huawei	Claims 40% transistor count reduction
High-performance ternary logic (CNT source-gating)	January 2025	<i>Science Advances</i>	Carbon nanotube implementation
Reconfigurable binary/ternary floating-gate devices	2025	<i>Nature Communications</i>	Heterojunction architecture

BitNet b1.58 (ternary LLM weights)	2024	Microsoft Research	Matches full-precision at 3× less energy
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The Church-Turing caveat: Computational complexity classes (P, NP, BQP) are invariant under change of base. The advantages of ternary are in constant factors and practical efficiency, not asymptotic complexity. This is important to state honestly — but constant factors are what determine whether a chip is commercially viable, whether a model fits in memory, whether a system runs on renewable energy. At engineering scale, constant factors are everything.

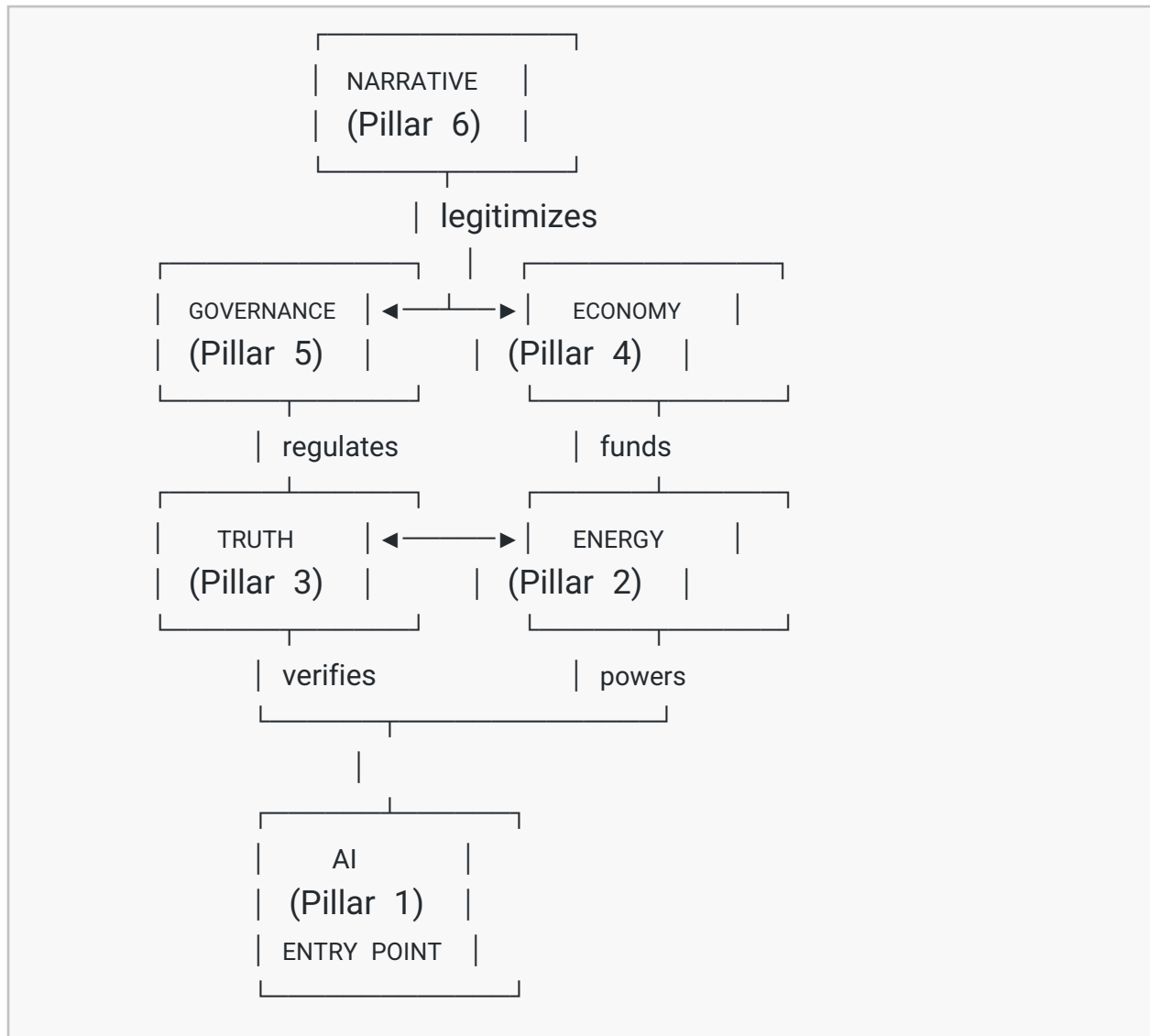
1.8 The Provably Optimal Classical Result

Bachet's weight problem (1612) provides the cleanest proof of balanced ternary's optimality: with a two-pan balance, weights of powers of 3 placed on either pan can measure any integer weight from 1 to $(3^n - 1)/2$ — a factor of approximately 1.5× better than binary powers on one pan. This is the only classical domain where balanced ternary has been proven strictly optimal, and it captures the essential insight: when you can use both signs (place weights on either pan), ternary beats binary by a provable margin.

Part II: Strategic Architecture — The Six Sovereign Pillars

The Flywheel Structure

The Parallel Empire is built on six interdependent pillars. They are not built simultaneously — they are activated in sequence, each one funding and enabling the next. The flywheel starts with Pillar 1 (AI) and cascades outward.



Pillar 1: Sovereign AI – The Entry Point

What it is: A ternary-native AI inference engine, starting with ternary-quantized open-weight models (Llama, Mistral, Qwen), progressing to a custom ternary training framework, and

ultimately to ternary-native foundation models.

Why it’s the entry point: AI is the highest-leverage technology of 2026. It touches every other pillar. A sovereign AI capability that is architecturally differentiated (ternary-native) and practically superior (lower energy, lower cost, faster inference) creates immediate economic value and attracts talent, capital, and attention.

Technical starting point (executable in 72 hours):

BitNet b1.58 has already proven the concept: ternary weights $\{-1, 0, +1\}$ match full-precision accuracy at 3× lower energy. The open-source implementations exist. The entry move is:

- 1. Fork the BitNet inference kernel (Microsoft’s 1-bit LLM framework, MIT licensed)
- 2. Implement balanced ternary arithmetic in the core matmul operations (replacing the current binary-encoded ternary)
- 3. Benchmark against binary baselines on standard tasks (MMLU, HumanEval, GSM8K)
- 4. Publish results openly, establishing technical credibility

The key architectural choice: Native balanced ternary representation in memory and arithmetic, not binary-encoded ternary. This means the weight tensor stores actual trits (using 2 bits per trit in current binary hardware, progressing to native trit storage as ternary hardware matures). The inference kernel performs balanced ternary addition and multiplication using the carry-free multiplication table and 2-carry addition table described in Part I.

Milestone targets:

Milestone	Timeline	Metric
Fork + balanced ternary matmul kernel	Week 1-2	Working code, passing unit tests
Benchmark on 7B parameter	Week 3-4	Perplexity within 1% of binary

model		baseline
Inference speed benchmark	Month 2	>2× throughput on consumer GPU
Energy benchmark	Month 2	>2.5× efficiency on consumer GPU
Public technical report	Month 3	Published on arXiv or equivalent
Community adoption (stars/forks)	Month 4–6	>1,000 GitHub stars
First external contributor	Month 3–6	Verified PR merged

Open-source starting points:

Component	Repository/Resource	License
BitNet inference	microsoft/BitNet (GitHub)	MIT
Base models for quantization	meta-llama/Llama-3, mistralai/Mistral	Llama Community / Apache 2.0
CUDA kernel development	NVIDIA/cutlass	BSD 3-Clause
Ternary arithmetic reference	Knuth, <i>TAOCP</i> Vol. 2, §4.1	Reference
Balanced ternary ALU design	Louis-dr/ternalu3 (GitHub)	Open

Pillar 2: Sovereign Energy

What it is: Distributed energy generation and storage that powers the AI infrastructure independently of centralized grids. Starting with solar+battery installations at compute nodes, progressing to a mesh energy network.

Why it connects: AI inference requires energy. Dependence on centralized energy grids is a sovereignty vulnerability. The ternary AI engine's 3× energy efficiency advantage means each watt goes further, making off-grid operation viable sooner.

Entry mechanism: The ternary AI inference engine's energy efficiency creates a measurable cost advantage. This advantage funds distributed solar+battery installations at edge compute locations. Each installation simultaneously serves as a proof-of-concept for the energy pillar and reduces operational cost for the AI pillar.

Precedent: Bitcoin mining drove massive investment in renewable energy infrastructure because miners follow cheap energy. The same dynamic applies here: ternary inference follows cheap, sovereign energy.

Phase 1 target (Month 6–12): One self-powered edge inference node. Solar array + battery + ternary inference server. Proof that the AI pillar can operate independently of grid energy.

Pillar 3: Sovereign Truth/Verification

What it is: A cryptographic verification layer that provides provable authenticity for AI outputs, data provenance, and identity — built on post-quantum cryptographic primitives that are structurally aligned with ternary arithmetic.

Why it connects: As AI-generated content becomes indistinguishable from human-generated content, the ability to verify provenance becomes critical infrastructure. The Eisenstein integer lattice and balanced ternary search space described in Part I are structurally aligned with NIST's post-quantum standards (Kyber, Dilithium).

Technical foundation: Lattice-based cryptography on the ternary lattice $\{-1, 0, 1\}^n$. Content signing using post-quantum schemes. Merkle tree commitments for data provenance. Zero-knowledge proofs for privacy-preserving verification.

Entry mechanism: Every output from the Pillar 1 AI engine is signed with a ternary-native post-quantum signature. This creates a growing corpus of verified AI outputs that demonstrates the verification layer's value.

Pillar 4: Sovereign Economy

What it is: An economic protocol that enables value exchange within the Parallel Empire without dependence on incumbent financial infrastructure. Not necessarily a cryptocurrency — possibly a mutual credit system, a time-based currency, or a programmable obligation network.

Why it connects: The AI engine generates economic value (inference-as-a-service). The energy network trades surplus capacity. The verification layer enables trust without intermediaries. These three create a natural economic substrate.

Design principles:

1. Earned, not mined (value comes from productive contribution, not proof-of-work)
2. Ternary-native accounting (balanced ternary arithmetic eliminates rounding errors in financial calculations because truncation = rounding)
3. Post-quantum secure (using Pillar 3's verification layer)
4. Programmable obligations (smart contracts for service agreements)

Precedent: The early Buddhist Sangha operated a complete parallel economy with its own rules of exchange, mutual support, and resource allocation — sustaining itself for 2,500+ years. Bitcoin demonstrated that a parallel monetary system can bootstrap from zero to trillion-dollar scale in 15 years.

Phase 1 target (Month 9–18): Internal credit system among founding team and early contributors. Track contributions, allocate credits, enable exchange.

Pillar 5: Sovereign Governance

What it is: Decision-making protocols for the Parallel Empire that are transparent, accountable, and resistant to capture. Not democracy-by-default — rather, a system designed from first principles for the specific requirements of a distributed technology civilization.

Design principles:

1. Consent-based (opt-in participation, exit rights preserved)
2. Competence-weighted (domain expertise matters for domain decisions)
3. Transparent process, private deliberation (decisions are auditable, but working discussions are protected)
4. Antifragile (stress makes the system stronger, not weaker)

Precedent: Linux kernel governance (Benevolent Dictator + lieutenants + meritocratic contribution), IETF rough consensus, Ostrom's commons governance principles (Nobel Prize 2009).

Phase 1 target (Month 12–24): Published governance charter. Decision log with public rationale. First contested decision resolved through the protocol.

Pillar 6: Sovereign Narrative

What it is: The memetic and cultural layer that makes the Parallel Empire legible, attractive, and self-reinforcing. The story that recruits builders, retains contributors, and provides meaning beyond economic incentive.

Why it's last but not least: Narrative without substance is propaganda. Substance without narrative is invisible. The narrative pillar is activated last because it requires real achievements to narrate — but it is designed from Day 1.

The core narrative (detailed in Part VI): *The universe computes in threes. Binary was a detour — necessary, productive, but architecturally suboptimal. The return to ternary is not regression but completion. We are building the infrastructure for the next computational era.*

Part III: Execution Roadmaps

Phase 0: Days 0–90 — The Tapas (Austerity/Heat)

This phase requires nothing but a computer, an internet connection, and disciplined execution. Zero capital expenditure. The tapas generates heat through concentrated effort.

Week 1–2: Foundation

Task	Deliverable	Verification
Set up development environment	Ubuntu/NixOS workstation with CUDA toolkit	nvidia-smi shows GPU
Fork BitNet repository	Personal GitHub repo with	Branch exists, CI passes

	balanced ternary branch	
Implement balanced ternary trit type	trit.h / trit.py with $\{-1, 0, 1\}$ arithmetic	Unit tests pass for all 9 add, 9 mul entries
Write balanced ternary matmul kernel (CPU)	bt_matmul.c or bt_matmul.py	Correctness verified against numpy
Begin technical blog/log	First post: “Why Balanced Ternary for AI Inference”	Published on personal site or GitHub Pages

Week 3–4: First Benchmark

Task	Deliverable	Verification
Quantize a 7B model to balanced ternary weights	Model checkpoint in BT format	File exists, loadable
Run perplexity evaluation on WikiText-103	Benchmark report	Perplexity within 2% of BF16 baseline
Profile memory usage	Memory report	>2× reduction vs BF16
Write second blog post	“Balanced Ternary 7B: First Benchmarks”	Published

Week 5–8: GPU Kernel

Task	Deliverable	Verification
Implement CUDA balanced ternary matmul	bt_matmul.cu	Correctness + >2× throughput vs BF16 on same GPU
Optimize memory layout for	Trit-packed tensor format	Memory benchmark shows

trit packing		expected compression
Run inference speed benchmarks	Tokens/second comparison	Published benchmark table
Open-source release v0.1	GitHub release with README, benchmarks, install instructions	Installable by external user in <10 minutes

Week 9–12: Community & Credibility

Task	Deliverable	Verification
Write arXiv-quality technical report	PDF with full methodology, benchmarks, ablations	Submitted to arXiv
Present at one online meetup/seminar	Recording or slides	Publicly accessible
Engage with BitNet/quantization community	Forum posts, issue responses, PR reviews	Observable activity
Reach 500+ GitHub stars	Star count	Visible on repo
Identify and recruit first contributor	Merged PR from external contributor	Git log
Energy efficiency benchmark	Joules per token comparison	Published data

Phase 0 Exit Criteria (all must be met)

- ☐ Working balanced ternary inference kernel (CPU + GPU)
- ☐ Benchmarked on ≥ 2 model sizes (7B, 13B)

- ☐ Perplexity within 1% of binary baseline
- ☐ Throughput $\geq 2\times$ binary baseline on same hardware
- ☐ Energy efficiency $\geq 2.5\times$ binary baseline
- ☐ Open-source release with documentation
- ☐ ≥ 500 GitHub stars
- ☐ ≥ 1 external contributor
- ☐ Technical report published
- ☐ ≥ 10 blog posts documenting the journey

Phase 1: Days 90–365 – The Churning

Capital may now enter. The technical credibility from Phase 0 attracts it. This phase builds the first real products and the first real community.

Quarter 2 (Months 4–6): Product

Objective	Deliverable	KPI
Launch inference-as-a-service API	REST API serving ternary-quantized models	<100ms p99 latency, >99.5% uptime
Build CLI tool for local ternary inference	asura-infer CLI	Installable via pip/cargo, <5 min setup
Implement model conversion pipeline	Binary $\rightarrow$ Balanced Ternary converter	Supports Llama, Mistral, Qwen families
Establish benchmark suite	Standardized eval harness for	Reproducible by any

	ternary models	researcher
Begin post-quantum signature integration	Signed inference outputs	Every API response includes BT-native PQ signature

Quarter 3 (Months 7–9): Scale

Objective	Deliverable	KPI
Training framework (Phase 1)	Ternary-aware training loop (STE + custom quantization)	Train 1B model from scratch in BT
Energy pilot	First solar-powered inference node	24-hour continuous operation demonstrated
Community growth	Active Discord/Matrix + monthly virtual meetup	>100 active weekly contributors
Developer documentation	Comprehensive docs site	>50 pages, tutorials for 5 common use cases
First revenue	Paying API customers	>\$1,000 MRR

Quarter 4 (Months 10–12): Consolidation

Objective	Deliverable	KPI
Ternary training at scale	Train 7B model natively in balanced ternary	Benchmark competitive with binary-trained 7B
Verification layer v1	Content provenance system using PQ signatures	Deployed on all API outputs
Economic protocol design	Whitepaper for internal credit	Published, peer-reviewed by

	system	≥3 external experts
Governance charter v1	Published decision-making framework	First 3 decisions made through the protocol
Narrative assets	Website, manifesto, visual identity, 3 long-form publications	Published and circulating
Year 1 retrospective	Public technical + strategic report	Published

Phase 1 Exit Criteria

- ☐ Ternary training framework operational
- ☐ ≥\$10,000 MRR from inference services
- ☐ ≥1 solar-powered inference node operational
- ☐ ≥500 active community members
- ☐ ≥10 external contributors to core codebase
- ☐ Post-quantum verification layer deployed
- ☐ Governance charter published and operational
- ☐ ≥3 peer-reviewed or arXiv publications
- ☐ Technical talk at ≥1 major conference (NeurIPS, ICML, or equivalent)

Phase 2: Years 2–5 — The Amrita (Nectar)

This is the compounding phase. Each pillar reinforces the others. The flywheel accelerates.

Year 2: Foundation Models + Energy Network

Pillar	Objective	Target
AI	Train 70B+ ternary-native foundation model	Top-10 on major benchmarks
AI	Ternary vision + multimodal models	Competitive with binary SOTA
Energy	10-node solar-powered inference mesh	Geographic distribution across ≥ 3 regions
Truth	Verification layer adopted by ≥ 5 external projects	Signed content volume >1M items
Economy	Internal credit system operational	>100 participants
Governance	First governance evolution (charter amendment)	Completed through protocol
Narrative	Brand recognition in AI research community	Invited talks, media coverage

Year 3: Ecosystem + Hardware

Pillar	Objective	Target
AI	Ternary model ecosystem (fine-tuning, adapters, agents)	>100 community-built models
Energy	Revenue from energy trading	>\$100K annual

	(surplus capacity)	
Truth	Verification protocol submitted for standardization	IETF or equivalent draft
Economy	External participants in credit system	>1,000
Hardware	First ternary FPGA/ASIC design partnership	Working prototype
Governance	Federated governance across multiple working groups	>5 active working groups

Year 4: Sovereignty Threshold

Pillar	Objective	Target
AI	Ternary models as default choice for efficiency-constrained deployment	>10% market share in edge AI
Energy	Energy network self-sustaining	Revenue > operating cost
Truth	Verification protocol in production use by ≥ 3 major platforms	>100M signed items
Economy	Credit system interoperable with external financial systems	Convertibility established
Hardware	First commercial ternary inference accelerator	Available for purchase
Governance	Governance model	Documented influence

	adopted/adapted by ≥ 3 external projects	
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Year 5: The Post-Indra Position

Pillar	Objective	Target
AI	Ternary-native training + inference as industry standard pathway	Recognized in surveys/reports
Energy	100+ node global inference-energy mesh	Operates across ≥ 10 countries
Truth	Verification layer as infrastructure (like TLS/HTTPS)	De facto standard for AI provenance
Economy	Functioning parallel economic layer	>10,000 participants, >\$10M annual volume
Hardware	Qutrit quantum interface operational	First qutrit-classical hybrid computation
Narrative	"Ternary computing" recognized as legitimate paradigm	Wikipedia article, textbook references

Part IV: Economic Model & Unit Economics

Revenue Model

The Parallel Empire generates revenue through four mechanisms, activated sequentially:

Mechanism 1: Inference-as-a-Service (Month 4+)

Unit economics:

Parameter	Binary Baseline	Ternary Target	Advantage
GPU memory per 7B model	~14 GB (BF16)	~4.5 GB (BT)	3.1×
Tokens/second (A100)	~120 tok/s	~300 tok/s	2.5×
Energy per 1M tokens	~0.5 kWh	~0.17 kWh	3×
Cost per 1M tokens (cloud GPU)	~\$0.60	~\$0.20	3×
Selling price per 1M tokens	—	\$0.30	50% margin

At 100M tokens/day (modest for a public API):

$$\text{Monthly revenue} = 100\text{M} \times 30 \times \$0.30/1\text{M} = \$900/\text{month}$$
$$\text{Monthly cost} = 100\text{M} \times 30 \times \$0.20/1\text{M} = \$600/\text{month}$$

Monthly margin = \$300/month (33%)

At 10B tokens/day (medium scale, ~Year 2):

Monthly revenue = \$90,000

Monthly cost = \$60,000

Monthly margin = \$30,000

Mechanism 2: Model Conversion Services (Month 6+)

Enterprises wanting to deploy existing models more efficiently pay for conversion to balanced ternary format plus ongoing support.

Tier	Service	Price
Self-serve	Open-source converter + docs	Free (community growth)
Supported	Conversion + validation + benchmark report	\$500–\$5,000 per model
Enterprise	Custom optimization + SLA + integration support	\$10,000–\$50,000/engagement

Mechanism 3: Energy Surplus Trading (Year 2+)

Solar-powered inference nodes generate more energy than needed during peak sunlight. Surplus is sold back to local grids or to other compute providers.

Mechanism 4: Verification-as-a-Service (Year 2+)

Content signing and provenance verification for AI-generated content. Per-signature pricing or subscription model.

Cost Structure

Phase 0 (Days 0–90): Near-Zero

Item	Cost	Funding Source
Cloud GPU (development/benchmarking)	\$200–\$500/month	Personal / free credits (Lambda, Colab)
Domain + hosting	\$20/month	Personal
Everything else	\$0	Sweat equity

Phase 1 (Days 90–365): Bootstrap

Item	Monthly Cost	Funding Source
Cloud GPU (production)	\$1,000–\$3,000	Revenue + grants
Solar pilot hardware	\$5,000–\$15,000 one-time	Grant or angel investment
Team (2–3 part-time contributors)	\$0–\$5,000 (equity + credits)	Internal credit system
Operations	\$500	Revenue

Break-Even Analysis

At \$0.30/1M tokens with 33% margin, break-even on \$3,000/month cloud GPU cost requires:

$$3,000 / 0.10 \text{ (margin per 1M tokens)} = 30,000\text{M} = 30\text{B tokens/month} \approx 1\text{B tokens/day}$$

This is achievable with approximately 50–100 active API users at moderate usage levels. Target: Month 8–10.

Part V: Recruitment & Signaling Playbook

Who You Need and When

Phase 0 (Solo Founder or Co-Founder Pair)

Profile: Systems programmer with CUDA experience OR machine learning engineer with quantization expertise. Ideally both in one person or a pair.

Where to find them: The person reading this document may be this person. If not:

Signal	Channel	Approach
BitNet/quantization contributors	GitHub activity	Direct message with specific technical question
Balanced ternary enthusiasts	r/math, Hacker News, math Twitter	Publish blog posts, let them find you
CUDA kernel developers	NVIDIA developer forums, GPU programming communities	Technical challenge: "Can you beat this benchmark?"
Former Setun researchers (yes, some are still active)	Academic networks, Russian language forums	Historical respect + modern relevance

Phase 1 (Core Team: 3–7 people)

Role	Why	When
ML Research Engineer	Training framework development	Month 4
Systems/Infrastructure Engineer	API, deployment, scaling	Month 5
Cryptographer	Post-quantum verification layer	Month 6
Technical Writer	Documentation, blog, arXiv papers	Month 4
Community Manager	Discord/Matrix, contributor onboarding	Month 6

Signaling Strategy: How to Attract Without Recruiting

The Parallel Empire does not hire. It attracts. The signaling strategy operates on three channels:

Channel 1: Technical Credibility (Primary)

Publish working code. Publish benchmarks. Publish honest assessments of what works and what doesn't. Technical credibility is the only signal that matters in Phase 0. Every blog post, every benchmark, every line of open-source code is a recruiting signal.

Channel 2: Narrative Resonance (Secondary)

The mythic frame is real and useful. It provides meaning beyond economic incentive. Builders who resonate with the narrative of building a parallel, sovereign computational civilization will self-select. Don't recruit them — let the narrative attract them.

Channel 3: Economic Alignment (Tertiary)

The internal credit system ensures that early contributors are compensated fairly when the economic pillar activates. Transparent, pre-published allocation rules. No stealth equity negotiations.

The Anti-Patterns to Avoid

Anti-Pattern	Why It Fails	What to Do Instead
Premature hiring	Burns cash, creates obligations before product-market fit	Attract contributors, compensate with credits
Stealth mode	Prevents community formation, no feedback loop	Build in public from Day 1
Grand announcements before results	Destroys credibility when results don't match	Publish results first, narrative second

Copying incumbent architecture	Competes on their terms, where they have advantages	Differentiate architecturally (ternary-native)
Adversarial framing	Creates enemies, invites retaliation	Parallel, not adversarial — “we’re building something new”

Part VI: Narrative & Memetic Framework

The Core Narrative

One-sentence version: *We are building the ternary computing layer — a sovereign, energy-efficient, mathematically optimal foundation for AI, cryptography, and computation that operates independently of the binary incumbent order.*

One-paragraph version: *Binary computing was a brilliant engineering choice for the vacuum tube era. It won because two-state switches were cheap and reliable, not because base 2 is mathematically optimal — base 3 is, by a provable margin. Now, as AI demands ever more efficient computation, as post-quantum cryptography requires lattice structures naturally aligned with ternary, and as qutrit quantum computing opens up richer Hilbert spaces, the mathematical arguments for ternary are converging with engineering feasibility. We are building the infrastructure for this transition: a ternary-native AI inference engine, a post-quantum verification layer, a sovereign energy mesh, and the economic and governance protocols to sustain them. We are not fighting the binary world. We are building the ternary one.*

Memetic Assets

The Key Phrases

Phrase	Function
"The universe computes in threes"	Orienting metaphor — ternary as natural
"Truncation is rounding"	Technical elegance made memorable (Knuth's observation)
"Negation is free"	Architectural advantage in one phrase
"Parallel, not adversarial"	Strategic posture — non-threatening, sovereign
"The third base"	Playful reference to Hayes' famous <i>American Scientist</i> article
"Post-binary"	Era-naming — positions ternary as the next paradigm
"Sovereignty through mathematics"	Core value proposition

The Visual Identity

The visual identity should be rooted in the mathematics:

The trit symbol: A three-state indicator. Three concentric elements: empty (0), filled-positive (+1), filled-negative (−1). Simple, reproducible, recognizable.

The hexagonal motif: Reflecting the Eisenstein integer lattice — the natural geometric home of ternary arithmetic in the complex plane. Hexagons tile the plane (like ternary digits tile the number line).

Color palette: Drawn from the three states: a neutral center (0), a warm positive (+1), a cool negative (−1). Suggested: charcoal/slate (0), amber/gold (+1), deep blue/indigo (−1).

Narrative Deployment Schedule

When	What	Where
Week 1	“Why I’m Building a Ternary AI Engine” (personal blog post)	Personal site, Hacker News, Twitter
Week 4	“First Benchmarks” (technical post with real data)	arXiv, GitHub, Reddit
Month 3	Technical report (full methodology)	arXiv
Month 6	“The Ternary Manifesto” (longer-form vision piece)	Personal site, invited publications
Month 9	Conference talk	NeurIPS/ICML workshop or equivalent
Month 12	Year 1 retrospective (honest assessment)	Personal site, major tech publications

Part VII: Risk Matrix & Pre-Mortem Mitigations

The Pre-Mortem Exercise

It is Month 12. The project has failed. Why?

Risk 1: “Ternary Doesn’t Actually Help” (Technical Risk — HIGH)

Scenario: Benchmarks show that balanced ternary inference is not meaningfully faster or more efficient than optimized binary quantization (GPTQ, AWQ, etc.).

Probability: 25%

Pre-mortem analysis: This would mean that the constant-factor advantages of ternary arithmetic are absorbed by the overhead of operating on binary hardware. The radix economy advantage is real but may be too small to overcome the optimization advantage of decades of binary infrastructure.

Mitigation:

1. Benchmark honestly and continuously. If ternary doesn’t win on current hardware, publish the negative result — this builds credibility.
2. Pivot focus to ternary training (where the advantages may be larger) rather than inference.
3. Accelerate the hardware partnership track (FPGAs/ASICs where ternary can be native).
4. Maintain the qutrit bridge — even if classical ternary advantages are marginal, the quantum

advantages are proven (70× gate count, 90× solution quality).

Trigger for pivot: If after 90 days of optimization, throughput advantage is <1.5× on same hardware.

Risk 2: “Incumbent Absorbs the Innovation” (Strategic Risk — MEDIUM)

Scenario: Microsoft, Google, or NVIDIA implements ternary quantization natively, capturing the efficiency gains within the binary ecosystem.

Probability: 40% (partial absorption is almost certain)

Pre-mortem analysis: BitNet b1.58 is already Microsoft Research. If they productize it, the inference efficiency advantage disappears for everyone.

Mitigation:

1. This is actually good for the mission. If ternary quantization becomes standard, it validates the thesis and normalizes the paradigm.
2. Differentiate on native ternary architecture (not just ternary weights on binary hardware) — this is harder to absorb.
3. Differentiate on the full stack (verification, energy, governance) — no incumbent wants to build all six pillars.
4. Move faster on the training framework, which is harder to replicate than inference.

Risk 3: “Nobody Cares” (Market/Adoption Risk —

MEDIUM)

Scenario: The technical results are good but nobody adopts them because the ecosystem (tools, libraries, models) is too small.

Probability: 30%

Mitigation:

1. Ensure 100% compatibility with existing model formats (convert in, convert out).
2. Make the efficiency gains so large they're worth the switching cost.
3. Build the narrative layer — technical results alone don't drive adoption; stories do.
4. Target a specific niche first (edge AI, where efficiency matters most) rather than competing broadly.

Risk 4: "Founder Burns Out" (Operational Risk — HIGH)

Scenario: The solo founder cannot sustain the intensity required across all dimensions (code, community, narrative, operations).

Probability: 35%

Mitigation:

1. Phase 0 is deliberately scoped for a single person working 40–60 hours/week, not 80+.
2. Build in public to create accountability and external motivation.
3. Recruit the first co-contributor by Month 3 at the latest.

4. Maintain non-project activities (exercise, relationships, rest). Tapas without recovery is self-destruction.
5. The 72-Hour Activation Sequence at the end of this document is designed to generate early momentum — momentum sustains motivation.

Risk 5: “Regulatory/Legal Barriers” (External Risk — LOW)

Scenario: Export controls, AI regulations, or patent claims block development or distribution.

Probability: 10%

Mitigation:

1. Open-source everything from Day 1 (hard to restrict what’s already public).
2. Use only permissively licensed dependencies.
3. Publish in academic venues (establishes prior art).
4. Distribute globally (no single jurisdiction can block a global open-source project — precedent: Linux, Bitcoin, Signal).

Full Risk Matrix

Risk	Probability	Impact	Mitigation Quality	Net Risk
Ternary doesn’t help	25%	Critical	Strong (pivot paths exist)	MEDIUM

Incumbent absorption	40%	High	Strong (validates thesis)	MEDIUM-LOW
No adoption	30%	High	Medium (requires execution)	MEDIUM
Founder burnout	35%	Critical	Medium (structural mitigations)	MEDIUM-HIGH
Regulatory barriers	10%	Medium	Strong (open-source, global)	LOW
Hardware never materializes	20%	Medium	Strong (software-first, qutrit bridge)	LOW
Better paradigm emerges	15%	High	Low (fundamental risk)	MEDIUM
Community toxicity/fragmentation	20%	Medium	Medium (governance charter)	LOW-MEDIUM

Part VIII: Weekly KPIs & Progress Proof

The Dashboard

Every Sunday, the founder(s) update these metrics. They are the proof that the tapas is generating heat.

Phase 0 KPIs (Weekly)

KPI	Target (Week 4)	Target (Week 8)	Target (Week 12)
Lines of code (core kernel)	2,000	5,000	10,000
Unit test coverage	80%	90%	95%
Benchmark perplexity gap vs binary	<5%	<2%	<1%
Throughput multiplier vs binary	1.5×	2.0×	2.5×
Energy efficiency multiplier	1.5×	2.5×	3.0×
GitHub stars	100	300	500+
Blog posts published	3	6	10
External contributors	0	0	1+
Technical report pages	0	5	20+
Hours of focused work this week	45+	45+	45+

Phase 1 KPIs (Monthly)

KPI	Month 4	Month 6	Month 9	Month 12
Monthly Recurring Revenue	\$0	\$500	\$3,000	\$10,000
API requests/day	1,000	50,000	500,000	5,000,000
Model formats supported	2	5	10	15+
Community members (active weekly)	20	50	100	200+
Core contributors	2	4	6	10+
Published papers	1	2	3	5+
Conference presentations	0	1	2	3+
Solar nodes operational	0	0	1	1+
Verified (signed) outputs	0	10K	100K	1M+

The Weekly Ritual

Every Sunday evening:

1. Update the KPI spreadsheet (15 minutes)
2. Write a brief internal log entry: what worked, what didn't, what changes (15 minutes)
3. Publish a brief external update: one thing learned, one result, one goal for next week (15 minutes)
4. Review the risk matrix: has any probability changed? (10 minutes)
5. Plan next week: three priorities, no more (10 minutes)

Total time: ~65 minutes/week. Non-negotiable.

72-Hour Activation Sequence

This Is What You Do Right Now

The following sequence is designed to generate irreversible forward momentum in 72 hours. Each step produces a tangible, visible artifact. By the end of Hour 72, you will have working code, a public identity, and a community signal — making it harder to stop than to continue.

Hour 0–4: Environment

- [] Install Ubuntu 24.04 or NixOS on your development machine (or verify existing setup)
- [] Install CUDA toolkit (if GPU available) or set up cloud GPU access (Lambda Labs free credits, Google Colab Pro)
- [] Install Python 3.11+, PyTorch 2.x, and Hugging Face Transformers
- [] Create a fresh GitHub repository: `[your-handle]/asura-ternary` (or your preferred name)
- [] Write a `README.md` with one paragraph: what this project is and why it exists
- [] Push to GitHub. The repository is now public. You have crossed the threshold.

Hour 4–12: First Code

- [] Implement `trit.py`: a balanced ternary digit type with the full arithmetic tables

```
# Starter template — balanced ternary trit type
class Trit:
    """A single balanced ternary digit: -1, 0, or +1."""
    __slots__ = ('value',)

    def __init__(self, v: int):
        assert v in (-1, 0, 1), f"Trit must be -1, 0, or 1, got {v}"
        self.value = v

    def __neg__(self):
        return Trit(-self.value) # Negation: swap -1 <-> 1, 0 stays

    def __mul__(self, other):
        return Trit(self.value * other.value) # Carry-free: always a single
trit
```

```
def __add__(self, other):
    s = self.value + other.value
    if s == 2:    return Trit(-1), Trit(1)    # carry +1
    if s == -2:  return Trit(1), Trit(-1)    # carry -1
    return Trit(s), Trit(0)                  # no carry
```

- [] Implement `balanced_ternary.py`: conversion between integers and balanced ternary representation
- [] Implement `bt_tensor.py`: a tensor class that stores weights as packed trits
- [] Write unit tests for all three modules
- [] Run tests. Green. Commit and push.

Hour 12–24: First Computation

- [] Implement `bt_matmul.py`: matrix multiplication using balanced ternary arithmetic
- [] Benchmark against numpy float32 matmul: correctness check on random matrices
- [] Load a small model (GPT-2 124M) and convert its weights to balanced ternary
- [] Run a single forward pass through one layer using your BT matmul
- [] Verify output matches (within tolerance) the float32 forward pass
- [] Commit and push with benchmark results in the commit message

Hour 24–36: First Content

- [] Write blog post #1: “Day 1: Building a Balanced Ternary AI Inference Engine”
 - What you built
 - Why balanced ternary (cite the mathematics from Part I of this Codex)

- First results (honest, including what doesn't work yet)
 - What's next
- ☐ Publish on your personal site, GitHub Pages, or Substack
- ☐ Post to Hacker News with a factual title (no hype)
- ☐ Post to r/MachineLearning with benchmarks
- ☐ Post to Twitter/X with one key result and a link

Hour 36–48: First Optimization

- ☐ Profile the BT matmul: where are the bottlenecks?
- ☐ Implement trit packing: store 5 trits per byte ($3^5 = 243 < 256$)
- ☐ Re-benchmark with packed representation
- ☐ If GPU available: write first CUDA kernel for BT matmul (even if slow)
- ☐ Update README with current benchmarks
- ☐ Commit and push

Hour 48–60: First Model

- ☐ Convert Llama-3 8B (or Mistral 7B) weights to balanced ternary
- ☐ Run perplexity evaluation on WikiText-103 (even if slow on CPU)
- ☐ Record: perplexity score, memory usage, tokens/second
- ☐ Compare to published BF16 and INT4 baselines
- ☐ Add results to README in a clear table
- ☐ Commit and push

Hour 60–72: First Signal

- ☐ Write blog post #2: “72 Hours In: Ternary 7B Benchmarks”
 - ☐ Create a CONTRIBUTING.md with clear instructions for how others can help
 - ☐ Create 5 GitHub Issues labeled “good first issue” for potential contributors
 - ☐ Create a ROADMAP.md linking to this Codex’s Phase 0 milestones
 - ☐ Set up a minimal community channel (GitHub Discussions, or a public Matrix room)
 - ☐ Update your GitHub profile to reflect this project
 - ☐ Send a brief update to 3–5 people whose opinion you respect, asking for honest technical feedback
 - ☐ Review the Phase 0 roadmap. Identify your three priorities for next week.
 - ☐ Commit. Push. Rest. The tapas has begun. The heat is real. The churn is underway.
-

Closing: The Honest Assessment

This Codex does not promise that building a parallel civilization layer is easy, likely, or even possible within five years. The mathematical foundations are sound — radix economy, carry-free multiplication, truncation-as-rounding, Eisenstein integer geometry, qutrit quantum advantages — but mathematics alone doesn’t build civilizations. Execution does. Community does. Sustained, disciplined effort over years does.

The threads documented in Part I — balanced ternary arithmetic, fractional part functions, transcendental constants, higher-dimensional geometry, and qutrit quantum computing — form a

constellation of related ideas, not a unified theory. The strongest connections (balanced ternary → symmetric residue → Hermite's identity → Cantor set → 3-adic integers) are algebraically tight. The connections to π are genuine but limited (π^2 has a base-3 BBP formula; π itself provably does not). The connections to higher geometry run through Eisenstein integers and lattice cryptography. The convergence point is qutrits, where ternary arithmetic meets quantum mechanics meets Lie algebra.

What this Codex provides is the highest-leverage entry point (ternary-native AI inference), the cascade logic (how each pillar enables the next), the execution roadmap (what to do, when, and how to measure progress), and the 72-hour activation sequence (how to start before doubt has time to settle).

The universe may or may not compute in threes. But the mathematics says that if you're going to compute, three states are provably closer to optimal than two. And in 2026, with ternary-weight LLMs matching full-precision accuracy at 3× lower energy, with post-quantum cryptography built on lattices naturally aligned with ternary structure, and with qutrit quantum computers demonstrating 90× advantages — the window to build the ternary layer is open.

The question is whether someone will walk through it.

"The Asura's tapas is not prayer. It is engineering. It is code committed and pushed. It is benchmarks published honestly. It is the patient, compounding construction of something real. The heat generated by this discipline is what reshapes the cosmos."

End of Codex v1.0

This document is itself an artifact of the Parallel Empire. Fork it. Improve it. Execute it.